

Sadie Lee

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Experience

Boston Children's Hospital

Jun 2026 – Present

Research Assistant – Computational Health Informatics Program

- Evaluating frontier LLMs for clinical decision support using clinician-annotated benchmark datasets to measure model safety and deployment readiness.
- Building scalable end-to-end data analytic pipelines in Python and R to ingest, clean, and analyze large Electronic Health Record (EHR) data.
- Engineered a deterministic NLP pipeline to parse unstructured clinical text, enabling targeted cohort segmentation and longitudinal treatment tracking.
- Conducting additional original research using neural network-informed differential equations to forecast long-term continuous glucose trajectories and dynamics.

Mayo Clinic

May 2025 – Aug 2025

Research Student – Mayo Clinic Platform

- Analyzed patient re-identification risks in vision transformers and convolutional neural networks to assess patient privacy and data leakage prior to exporting model weights outside secure environments.
- Executed model inversion attacks using a generative adversarial network to reconstruct medical images and predict demographic attributes from frozen model weights.

Voythos

May 2025 – Jul 2025

Data Science Intern

- Developed multi-task vision transformers on volumetric chest CT images for aortic segmentation and thrombosis detection, feeding downstream multimodal models to predict risk of Type B aortic dissection.
- Refactored machine learning pipelines and optimized data processing, resolving execution bugs and accelerating model training time by 97%.

Mayo Clinic

May 2024 – Aug 2024

Undergraduate Intern – Mayo Clinic Platform

- Built dashboards in Power BI to geographically track customer acquisitions, visualize account metrics, and present product KPIs.
- Established an internal pipeline to support health tech start-up customers in market readiness with access to de-identified medical images for AI/ML solution development.
- Trained 3D convolutional neural networks in PyTorch for abnormality classification and segmentation.
- Tested model training performance across NVIDIA GPUs to evaluate infrastructure and resource constraints.
- Queried relational clinical databases using SQL to assess data availability and scope feasibility for prospective customers.

BC Children's Hospital Research Institute, UBC Faculty of Medicine

May 2023 – Apr 2024

Research Assistant

- Evaluated the impact of a digital peer support intervention on glycemic control in adults with type 1 diabetes as part of an interventional clinical trial.
- Coded R scripts to ingest and process glucose sensor time-series data, calculating changes in sensor-derived metrics pre- and post-intervention.
- Served as the primary liaison between software engineering, research, and clinical teams to manage app deployment, document bug fixes, and lead user onboarding.
- Co-authored study protocols, Data Safety and Monitoring Board (DSMB) plans, and ethics amendments.

Education

University of British Columbia, Vancouver, Canada

May 2026

Bachelor of Arts in Cognitive Systems, Minor in Data Science – High Distinction

Relevant coursework: Statistical Modeling, Databases, Applied Machine Learning, Data Structures and Algorithms, Data Visualization, Discrete Math

Honors and Awards

Andrew Wade Memorial Award in Visual Analytics

Mar 2026

UBC Arts Undergraduate Society Student Conference Grant Award

Mar 2025

Technical Skills

Languages: Python, R, SQL, JavaScript/TypeScript, HTML/CSS

Tools/Frameworks: PyTorch, Git, Docker, Linux/Bash, MongoDB, Oracle, Power BI, React, REDCap

Research

T1D REACHOUT–A Mobile App to Deliver Peer-Led Mental Health Support to Adults Living With Type 1 Diabetes: Co-Design and Development Process
Sadie Lee, Baray Sidhu, Parteek Johal, Ayman Azhar, Jonath Sujan, Matthias Görges, Tricia S Tang
JMIR Diabetes. Mar 2026.

Re-identification Risk of Medical Imaging-Based Deep Learning Models
Sadie Lee; Supervised by Adam Resnick and Rob Blundo (Mayo Clinic Platform)
Undergraduate Capstone. Aug 2025.

Formalizing Ethical Design in Prostate Cancer Image Analysis: A Preliminary Case Study
Sadie Lee, Adam Resnick, Nasibeh Zanjirani Farahani
IEEE MIT Undergraduate Research Technology Conference. Oct 2024.

Topological Data Analysis and Interpretability of 3D-Convolutional Neural Networks
Sadie Lee
AAAI Undergraduate Consortium. Feb 2024.

Teaching and Volunteering

University of British Columbia

Jan – Apr 2026

Teaching Assistant - COGS 401 (Seminar in Cognitive Systems)

UBC Multidisciplinary Undergraduate Research Conference

Mar 2026

Adjudicator

UBC Emerging Media Lab

May – Aug 2023

Software Developer

UBC Multifaceted Innovations in Neurotechnology

Mar – Dec 2023

Deep Learning Researcher

UBC 180 Degrees Consulting

Jan – Apr 2023

Data Science Consultant